Application Based Learning – Individual Task

Module 4: Deep Learning

Topic: Artificial Neural Network (ANN) Flowchart

Introduction

Artificial Neural Networks (ANN) are one of the core concepts of Deep Learning and are inspired by the structure and functioning of the human brain. Just like biological neurons, artificial neurons are connected to each other and work together to process information and solve complex problems.

An ANN consists of multiple layers of interconnected nodes, including the input layer, hidden layers, and output layer. Each neuron receives input, processes it using weights and bias, and produces an output through an activation function. These networks are capable of learning patterns from data and making predictions without being explicitly programmed.

Artificial Neural Networks are widely used in real-world applications such as image recognition, speech recognition, natural language processing, recommendation systems, and medical diagnosis. They are especially powerful in handling large and complex datasets where traditional algorithms may not perform well.

One of the key features of ANN is its ability to learn from data through a process called training. During training, the network adjusts its internal parameters (weights and biases) to minimize errors and improve accuracy. This learning process is achieved using algorithms such as backpropagation, which helps the network understand and correct its mistakes.

Thus, Artificial Neural Networks play a crucial role in modern Artificial Intelligence and Deep Learning systems, enabling machines to perform tasks that require human-like intelligence.

Structure of Artificial Neural Network

An Artificial Neural Network (ANN) is composed of multiple layers of neurons that work together to process information. These layers are mainly divided into three types: Input Layer, Hidden Layer, and Output Layer.

Input Layer

The input layer is the first layer of the neural network. It receives the raw data from the external environment and passes it to the next layer. Each neuron in this layer represents a feature of the input data. For example, in image processing, each pixel value can act as an input.

The input layer does not perform any complex computations; it only transfers the data to the hidden layers.

Hidden Layer

The hidden layer is the most important part of the neural network. It performs all the computations and extracts patterns from the data. A network can have one or more hidden layers depending on the complexity of the problem.

Each neuron in the hidden layer receives input values, multiplies them with weights, adds a bias, and applies an activation function to produce an output. These operations allow the network to learn complex relationships within the data.

Output Layer

The output layer is the final layer of the neural network. It produces the final result or prediction based on the processed data.

The number of neurons in this layer depends on the type of problem. For example:

- One neuron for binary classification
- Multiple neurons for multi-class classification

Example:

The following diagram represents the structure of an Artificial Neural Network:

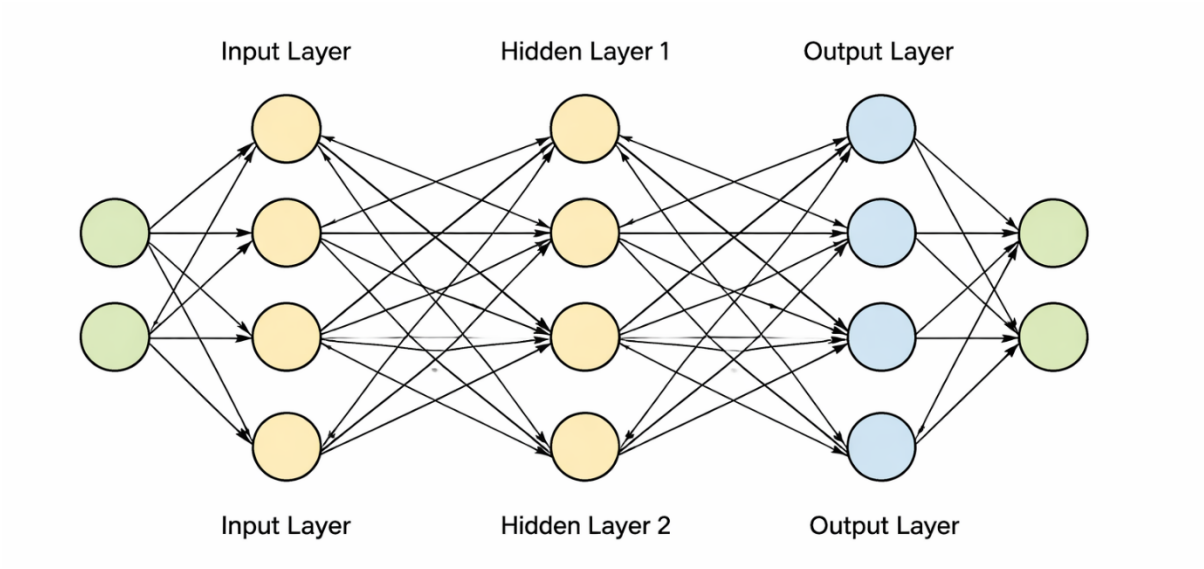

Figure: Basic Structure of Artificial Neural Network

Activation Functions in ANN

Activation functions play a crucial role in Artificial Neural Networks. They determine whether a neuron should be activated or not by transforming the input signal into an output signal.

Without activation functions, a neural network would behave like a simple linear model and would not be able to learn complex patterns. Activation functions introduce non-linearity into the network, allowing it to solve complex problems.

Types of Activation Functions

1. Sigmoid Function

The sigmoid function converts input values into a range between 0 and 1. It is commonly used in binary classification problems.

Formula:

$\sigma(x) = 1 / (1 + e^{-x})$

It is smooth and easy to understand but suffers from the vanishing gradient problem.

2. ReLU (Rectified Linear Unit)

ReLU is one of the most widely used activation functions. It outputs zero for negative values and returns the input value for positive values.

Formula:

$f(x) = \max(0, x)$

ReLU is efficient and helps in faster training of deep neural networks.

3. Tanh Function

The tanh (hyperbolic tangent) function converts input values into a range between -1 and 1.

Formula:

$\tanh(x) = (e^{x} - e^{-x}) / (e^{x} + e^{-x})$

It is zero-centered and performs better than sigmoid in many cases, but still faces vanishing gradient issues.

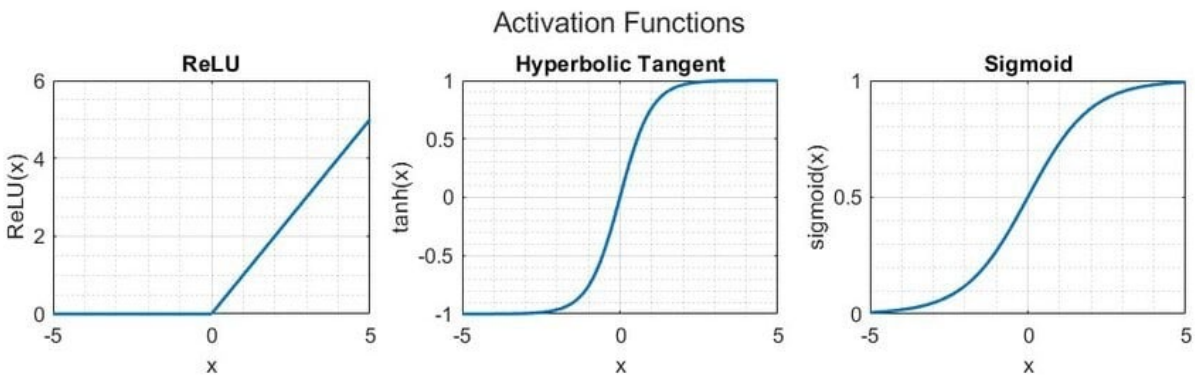

Figure: Sigmoid, ReLU, and Tanh Activation Functions

Backpropagation in Artificial Neural Networks

Backpropagation is a key algorithm used to train Artificial Neural Networks. It is used to minimize the error by adjusting the weights and biases of the network.

The main goal of backpropagation is to reduce the difference between the predicted output and the actual output.

How Backpropagation Works

The process of backpropagation involves the following steps:

1. **Forward Pass**
   In this step, the input data is passed through the network layer by layer to generate an output.
2. **Error Calculation**
   The output produced by the network is compared with the actual output to calculate the error.
3. **Backward Pass**
   The error is propagated backward through the network from the output layer to the input layer.
4. **Weight Update**
   The weights are updated using optimization techniques such as gradient descent to minimize the error.

Weight Update Formula

The weights are updated using the formula:

$$W(new) = W(old) - \text{learning rate} \times \text{gradient of error}$$

This means:

- If error is high → weights change more
- If error is low → small adjustments

Importance of Backpropagation

- Helps the network learn from mistakes
- Improves prediction accuracy
- Enables deep learning models to perform complex tasks

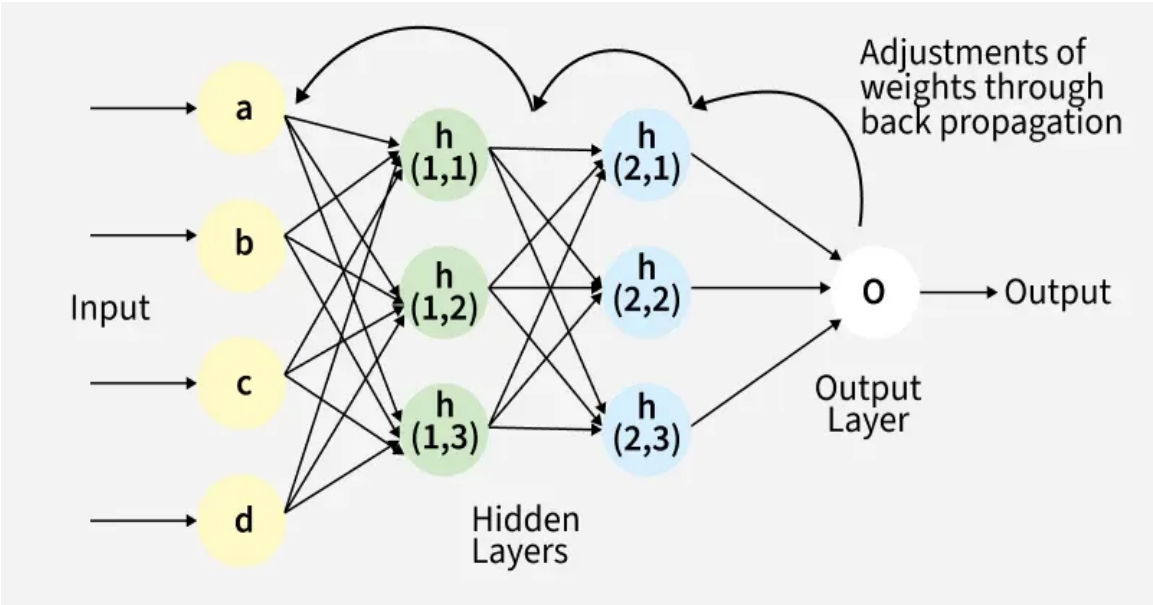

Figure: Backpropagation Process in Neural Network

Conclusion

Artificial Neural Networks (ANN) are a powerful component of deep learning that enable machines to learn patterns and make intelligent decisions. By organizing neurons into input, hidden, and output layers, ANN can process complex data and solve real-world problems effectively.

Activation functions such as Sigmoid, ReLU, and Tanh play an important role in introducing non-linearity, allowing the network to learn complex relationships. Backpropagation further enhances the learning process by continuously adjusting weights and reducing errors.

Through this assignment, we understand how ANN works internally, from data input to prediction output, and how learning is achieved through weight updates. ANN has wide applications in areas such as image recognition, speech processing, and healthcare, making it an essential concept in Artificial Intelligence and Machine Learning.